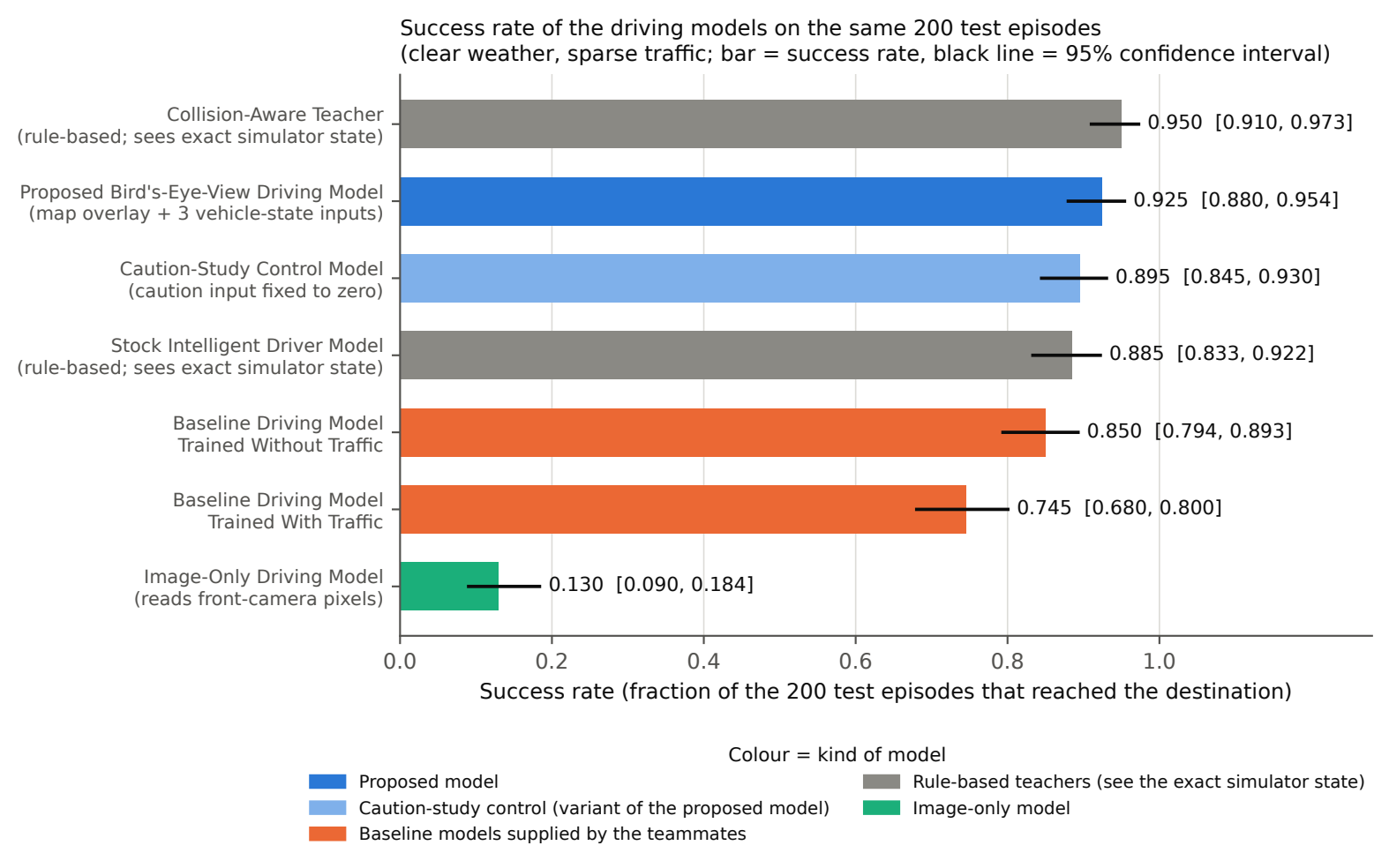
Success rate of the driving models on the same 200 test episodes
(clear weather, sparse traffic; bar = success rate, black line = 95% confidence interval)
Collision-Aware Teacher
(rule-based; sees exact simulator state)
0.950 [0.910, 0.973]
Proposed Bird's-Eye-View Driving Model
(map overlay + 3 vehicle-state inputs)
0.925 [0.880, 0.954]
Caution-Study Control Model
(caution input fixed to zero)
0.895 [0.845, 0.930]
Stock Intelligent Driver Model
(rule-based; sees exact simulator state)
0.885 [0.833, 0.922]
Baseline Driving Model
Trained Without Traffic
0.850 [0.794, 0.893]
Baseline Driving Model
Trained With Traffic
0.745 [0.680, 0.800]
Image-Only Driving Model
(reads front-camera pixels)
0.130 [0.090, 0.184]
0.0
0.2
0.4
0.6
0.8
1.0
Success rate (fraction of the 200 test episodes that reached the destination)
Colour = kind of model
Proposed model
Caution-study control (variant of the proposed model)
Baseline models supplied by the teammates
Rule-based teachers (see the exact simulator state)
Image-only model